# Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.

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# Project Overview & Dataset Summary

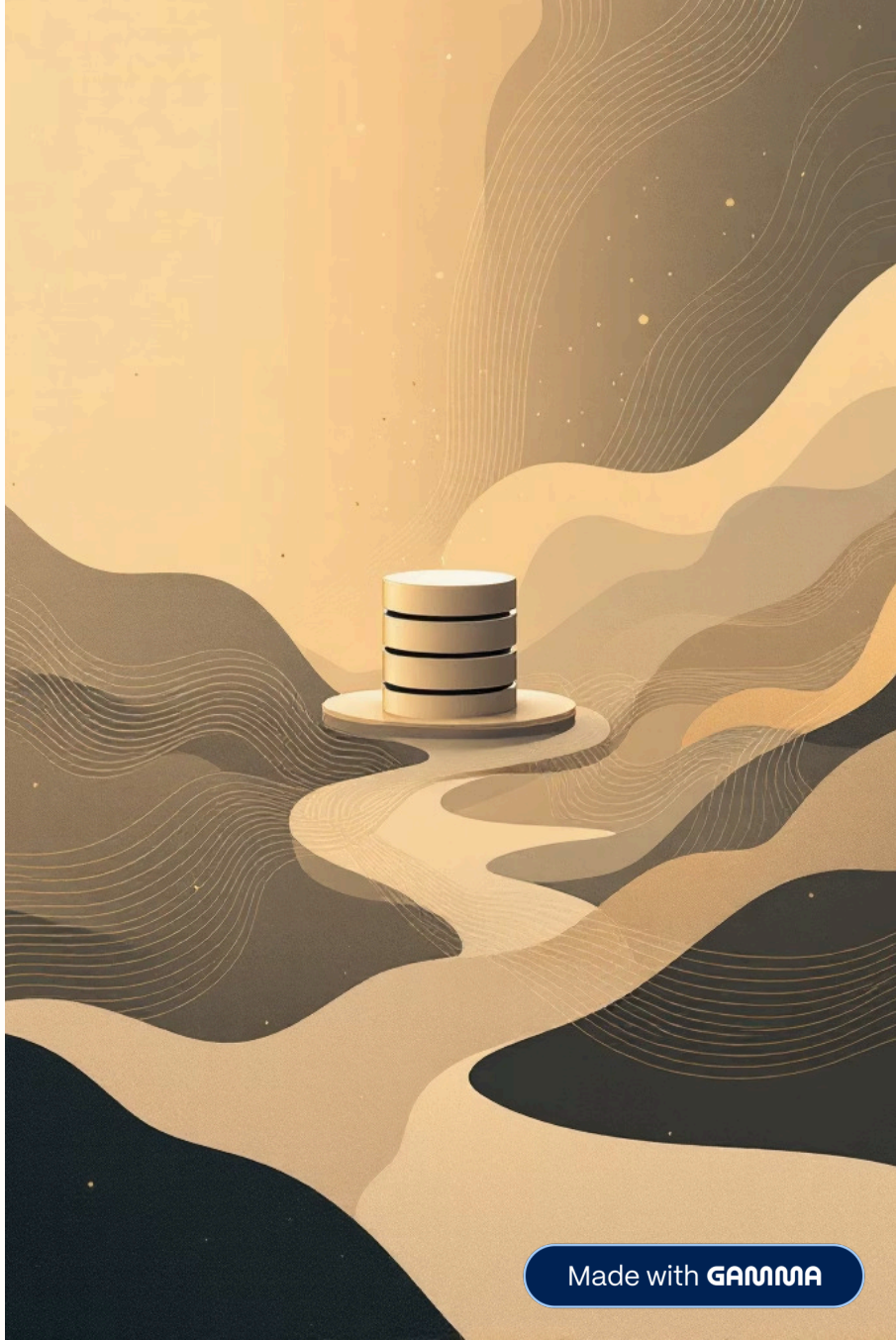
## Project Goal

Analyze spending patterns, segments, preferences, and subscription behavior.

Optimize operations and inform strategy.

## Dataset Snapshot

- 3,900 Rows, 18 Columns
- Key Features: Demographics, Purchase Details, Behavior
- Missing Data: 37 values in Review Rating



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# Exploratory Data Analysis (Python)

01

## Data Loading & Exploration

Imported with pandas, checked structure and summary statistics.

02

## Data Cleaning

Imputed missing Review Ratings (median by category).

03

## Column Standardization

Renamed columns to snake_case for readability.

04

## Feature Engineering

Created age_group and purchase_frequency_days.

05

## Data Consistency

Dropped redundant promo_code_used column.

06

## Database Integration

Loaded cleaned data into MySQL for SQL analysis.

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# Key SQL Analysis Findings

## Revenue by Gender

Male: $157,890

Female: $75,191

## Top 5 Products by Rating

- Gloves: 3.86
- Sandals: 3.84
- Boots: 3.82
- Hat: 3.80
- Skirt: 3.78

## Shipping Type Comparison

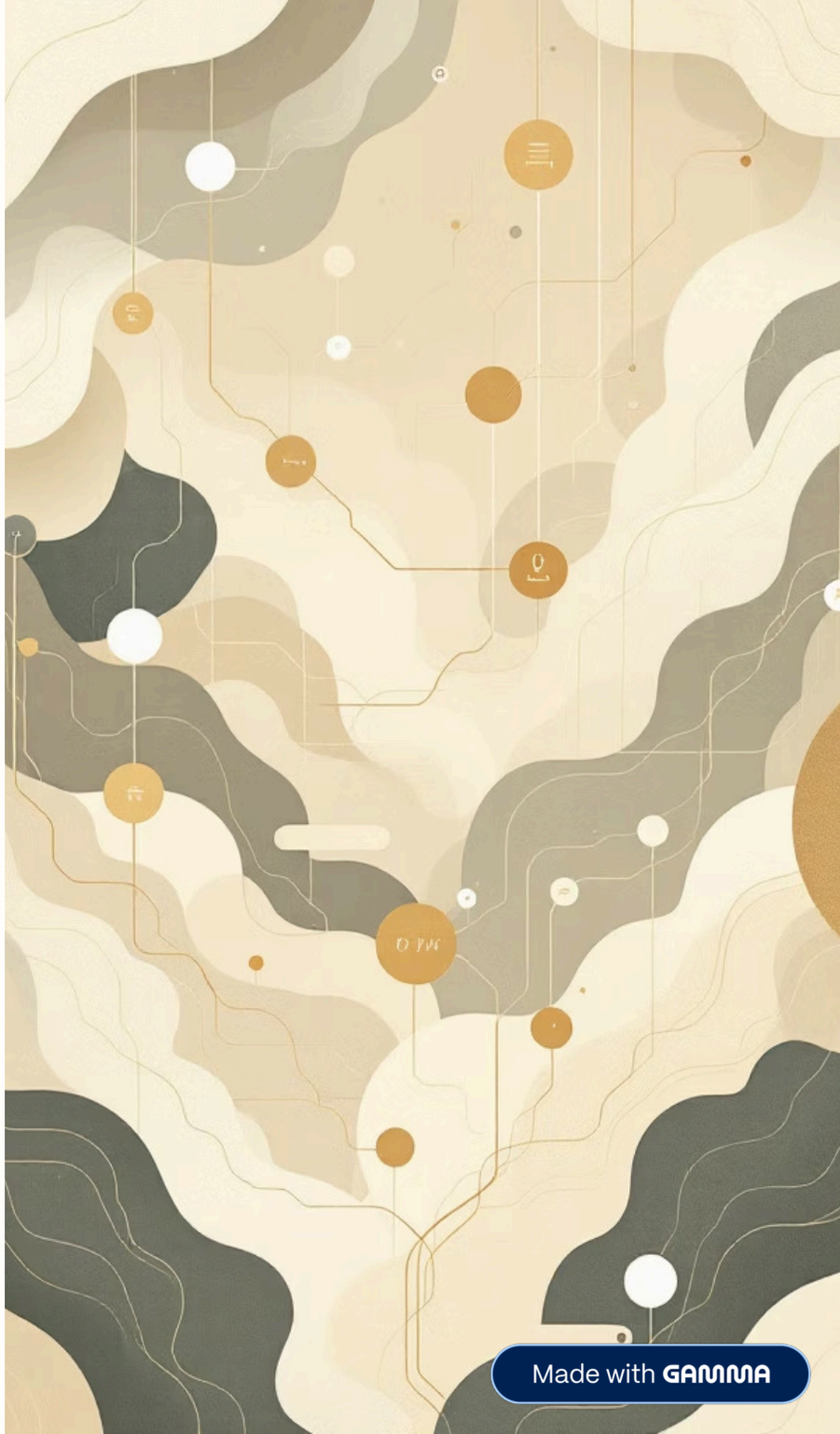
Standard: $58.46 avg.

Express: $60.48 avg.

## Subscribers vs. Non-Subscribers

Subscribers: 1053 customers, $59.49 avg. spend, $62,645 total revenue

Non-Subscribers: 2847 customers, $59.87 avg. spend, $170,436 total revenue



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# SQL Insights: Discounts & Segments

## Discount-Dependent Products

- Hat: 50.00%
- Sneakers: 49.66%
- Coat: 49.07%
- Sweater: 48.17%
- Pants: 47.37%

## Customer Segmentation

- Loyal: 3116
- Returning: 701
- New: 83

## Repeat Buyers & Subscriptions

Customers with >5 purchases:

No Subscription: 2518

Yes Subscription: 958

## Revenue by Age Group

- Young Adult: $62,143
- Middle-aged: $59,197
- Adult: $55,978
- Senior: $55,763

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# Power BI Dashboard: Visualizing Insights

Interactive dashboard for visual data exploration and presentation.

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# Key Business Recommendations

### Boost Subscriptions

Promote exclusive benefits for subscribers.

### Customer Loyalty Programs

Reward repeat buyers to foster loyalty.

### Review Discount Policy

Balance sales boosts with margin control.

### Product Positioning

Highlight top-rated and best-selling products.

### Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

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